

Chapter 9

Inverse Trigonometric Functions

Only One Option Correct Type Questions

1. If $0 < x < 1$, then $\sqrt{1+x^2} \cdot [x \cos(\cot^{-1} x) + \sin(\cot^{-1} x)]^2 - 1]^{\frac{1}{2}} =$ [IIT-JEE-2008 (Paper-1)]
- (A) $\frac{x}{\sqrt{1+x^2}}$ (B) x
- (C) $x\sqrt{1+x^2}$ (D) $\sqrt{1+x^2}$
2. The value of $\cot\left(\sum_{n=1}^{23} \cot^{-1}\left(1 + \sum_{k=1}^n 2k\right)\right)$ is [JEE (Adv)-2013 (Paper-1)]
- (A) $\frac{23}{25}$ (B) $\frac{25}{23}$
- (C) $\frac{23}{24}$ (D) $\frac{24}{23}$

One or More Option(s) Correct Type Questions

3. If $\alpha = 3 \sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3 \cos^{-1}\left(\frac{4}{9}\right)$, where the inverse trigonometric functions take only the principal values, then the correct option(s) is (are) [JEE (Adv)-2015 (Paper-2)]
- (A) $\cos\beta > 0$ (B) $\sin\beta < 0$
- (C) $\cos(\alpha + \beta) > 0$ (D) $\cos\alpha < 0$
4. For any positive integer n , define $f_n : (0, \infty) \rightarrow \mathbb{R}$ as $f_n(x) = \sum_{j=1}^n \tan^{-1}\left(\frac{1}{1+(x+j)(x+j-1)}\right)$ for all $x \in (0, \infty)$

(Here, the inverse trigonometric function $\tan^{-1}x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.)

Then, which of the following statement(s) is (are) TRUE?

[JEE (Adv)-2018 (Paper-2)]

- (A) $\sum_{j=1}^5 \tan^2(f_j(0)) = 55$
- (B) $\sum_{j=1}^{10} (1 + f'_j(0)) \sec^2(f_j(0)) = 10$
- (C) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \tan(f_n(x)) = \frac{1}{n}$
- (D) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \sec^2(f_n(x)) = 1$

5. For any positive integer n , let $S_n : (0, \infty) \rightarrow \mathbb{R}$ be defined by

$$S_n(x) = \sum_{k=1}^n \cot^{-1} \left(\frac{1 + k(k+1)x^2}{x} \right)$$

where for any $x \in \mathbb{R}$, $\cot^{-1}(x) \in (0, \pi)$ and $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then which of the following statements is

(are) **TRUE?**

[JEE (Adv)-2021 (Paper-1)]

(A) $S_{10}(x) = \frac{\pi}{2} - \tan^{-1} \left(\frac{1 + 11x^2}{10x} \right)$, for all $x > 0$

(B) $\lim_{n \rightarrow \infty} \cot(S_n(x)) = x$, for all $x > 0$

(C) The equation $S_3(x) = \frac{\pi}{4}$ has a root in $(0, \infty)$

(D) $\tan(S_n(x)) \leq \frac{1}{2}$, for all $n \geq 1$ and $x > 0$

Matrix-Match / Matrix Based Type Questions

6. Let (x, y) be such that

$$\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}$$

Match the statements in **Column-I** with statements in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

[IIT-JEE-2007 (Paper-2)]

Column-I

- (A) If $a = 1$ and $b = 0$, then (x, y)
 (B) If $a = 1$ and $b = 1$, then (x, y)
 (C) If $a = 1$ and $b = 2$, then (x, y)
 (D) If $a = 2$ and $b = 2$, then (x, y)

Column-II

- (p) lies on the circle $x^2 + y^2 = 1$
 (q) lies on $(x^2 - 1)(y^2 - 1) = 0$
 (r) lies on $y = x$
 (s) lies on $(4x^2 - 1)(y^2 - 1) = 0$

Integer / Numerical Value Type Questions

7. Let $f(\theta) = \sin \left(\tan^{-1} \left(\frac{\sin \theta}{\sqrt{\cos 2\theta}} \right) \right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. Then the value of $\frac{d}{d(\tan \theta)} (f(\theta))$ is

[IIT-JEE-2011 (Paper-1)]

8. Let $f : [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x) = \frac{10-x}{10}$ is

[JEE (Adv)-2014 (Paper-1)]

9. The number of real solutions of the equation $\sin^{-1} \left(\sum_{i=1}^{\infty} x^{i+1} - x \sum_{i=1}^{\infty} \left(\frac{x}{2} \right)^i \right) = \frac{\pi}{2} - \cos^{-1} \left(\sum_{i=1}^{\infty} \left(-\frac{x}{2} \right)^i - \sum_{i=1}^{\infty} (-x)^i \right)$ lying in the interval $\left(-\frac{1}{2}, \frac{1}{2}\right)$ is _____.

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\cos^{-1}x$ assume values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $[0, \pi]$, respectively.)

[JEE (Adv)-2018 (Paper-1)]

10. The value of $\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)\right)$ in the interval $\left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]$ equals ____.

[JEE (Adv)-2019 (Paper-2)]

11. Considering only the principal values of the inverse trigonometric functions, the value of

$$\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}+\frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2}+\tan^{-1}\frac{\sqrt{2}}{\pi} \text{ is } \underline{\hspace{2cm}}.$$

[JEE (Adv)-2022 (Paper-1)]

